AGENTIC AI HACKATHON 2026 · BUSINESS & PRODUCTIVITY AI

Business Listing Consistency Agent

One true record. Five listing sites. An agent that finds every mismatch, fixes only what you approve, and proves the fix went live.



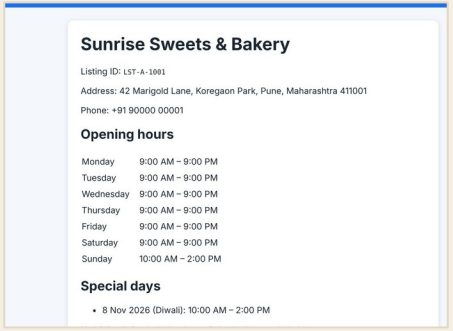
Live demo zirohacksub.onrender.com **demo PIN** 7KQ4X-M9PHT **Repo** github.com/yourgrok/zirohacksub

Fictional business (Sunrise Sweets & Bakery, Pune), mock listing sites, synthetic data only.

One bakery, five listing sites, five different stories

A listing inconsistency is any difference between what the owner means and what a site shows: name, address, phone or hours.

A · Listify



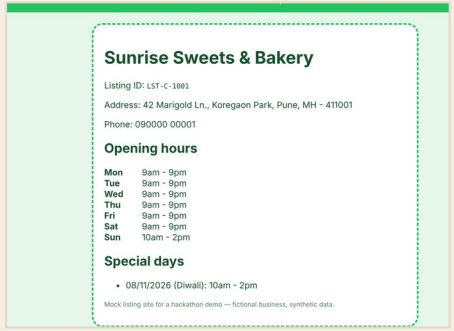
Matches the record

B · CityPages



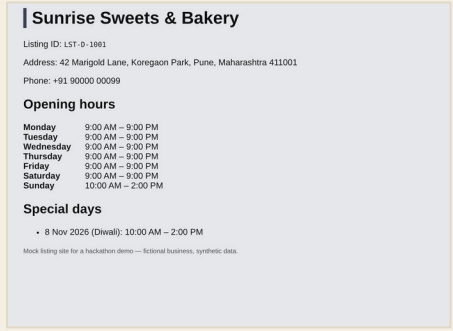
Diwali shown as Closed (record: 10:00-2:00)

C · LocalLoop



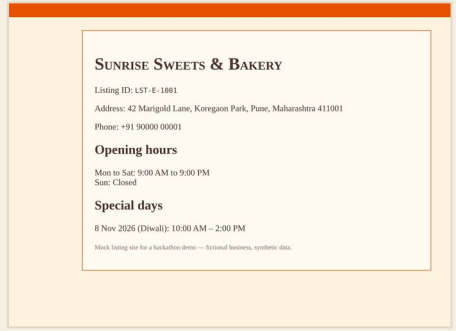
Phone and address written differently, same meaning

D · NearbyNow



Other phone number; claimed by another account

E · ShopFront



Sunday shown as Closed (record: 10:00-2:00)

Customers are turned away

Wrong hours or a stale phone number cost real orders.

Every site is different

Own formats and logins. Some listings are claimed by others, and some edits wait for a moderator.

Nobody checks afterwards

Drift returns silently. And a bot that edits freely is worse than no bot.

Tell us the truth once. The agent keeps every site honest.

1 — Scan

Reads every site and compares meaning, not formatting.

2 — Review

You approve or reject each fix. The reason is shown.

3 — Apply

Only approved changes are sent, never to a listing you do not own.

4 — Verify

Reloads the page. VERIFIED only if the value is really there.



Who it helps

- Busy owners of local shops
- Owners with more than one business
- New owners: found, and added where missing

A goal-driven loop, not a chat box

Goal every listing says what the owner's record says. The owner approves; the agent proves it.

1 Perceive

Read every page and dashboard. Keep the raw values.

2 Decide

Compare meaning. Pick the next tool from the current state.

3 Act

Propose, flag, ask the owner, or apply what was approved.

4 Verify

Reload the page. VERIFIED only if the value is really there.

5 Recover

Page changed? Site down? Re-ask, retry, re-send once.

loops until nothing is left to do

Eight tools, one registry

`read_listing`	`normalise`	`compare`	`propose`
`apply_approved`	`reload_and_verify`	`flag_listing`	`ask_owner`

Who decides each step?

A rules planner by default, or optional Gemini. Same tools, same guardrails. Every trace row shows who decided.

From plain-language request to verified result

Interprets requests

- Typed text becomes an intent: question, edit or date
- 26th December becomes Sat 26 Dec 2026
- Unclear? One question. Nothing saves before Confirm

"closed on 26th December" > Action: Closed, Date: Sat 26 Dec 2026

Selects tools

- Looks at every site's state before each step
- Read, normalise, compare, then propose
- Not your listing? Code forces a flag, never an edit

Rogue planner test: 6 unsafe picks, 6 blocked, 0 writes

Runs multi-step work

- Scan reads and proposes; Execute applies and verifies
- One closure request fans out to all five sites
- Every step is a visible trace row

TC1: MISMATCH > PROPOSED > APPROVED > APPLIED > VERIFIED

Verifies and recovers

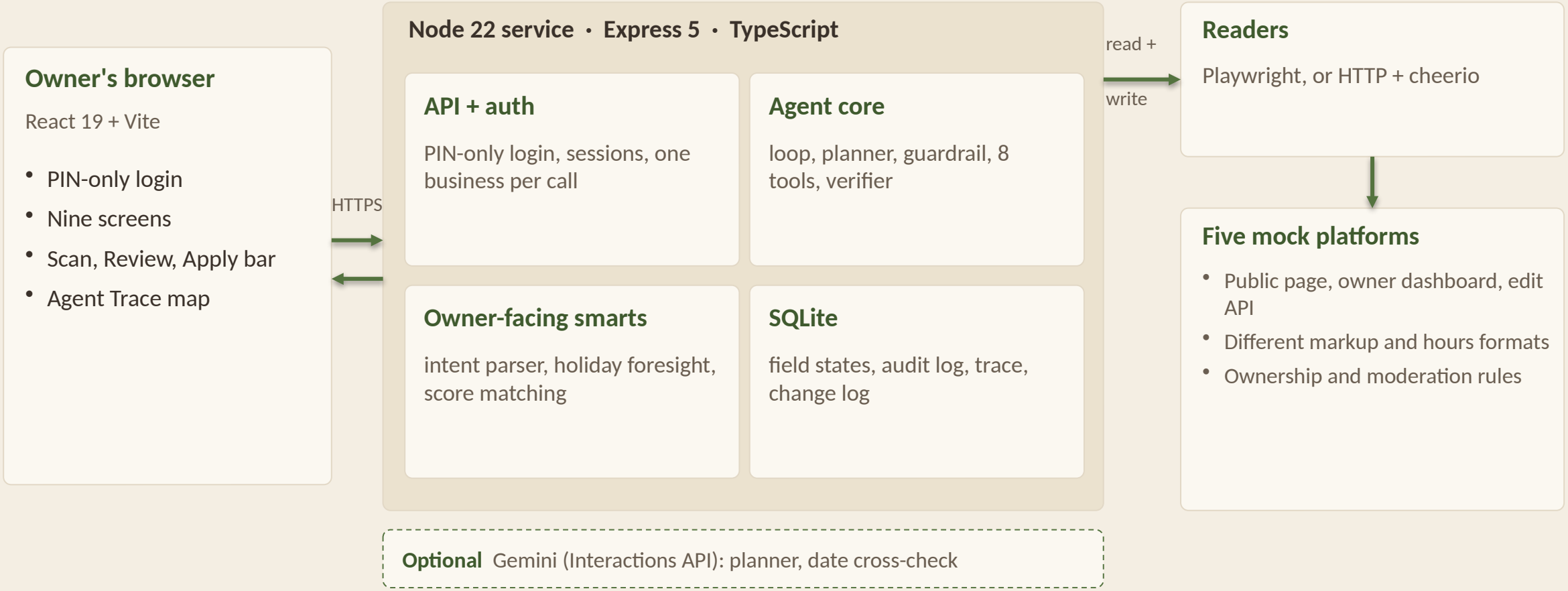
- Reloads the page; "saved" is never trusted
- A moderated site stays PENDING until it really changes
- Page changed? Site down? Re-ask, retry, re-send once

TC8: outage retried, lost write re-sent, VERIFIED

Honest note The default decision-maker is a rules planner. Gemini is optional, falls back to rules, and is tested with a mocked client: no live Gemini run is in our evidence.

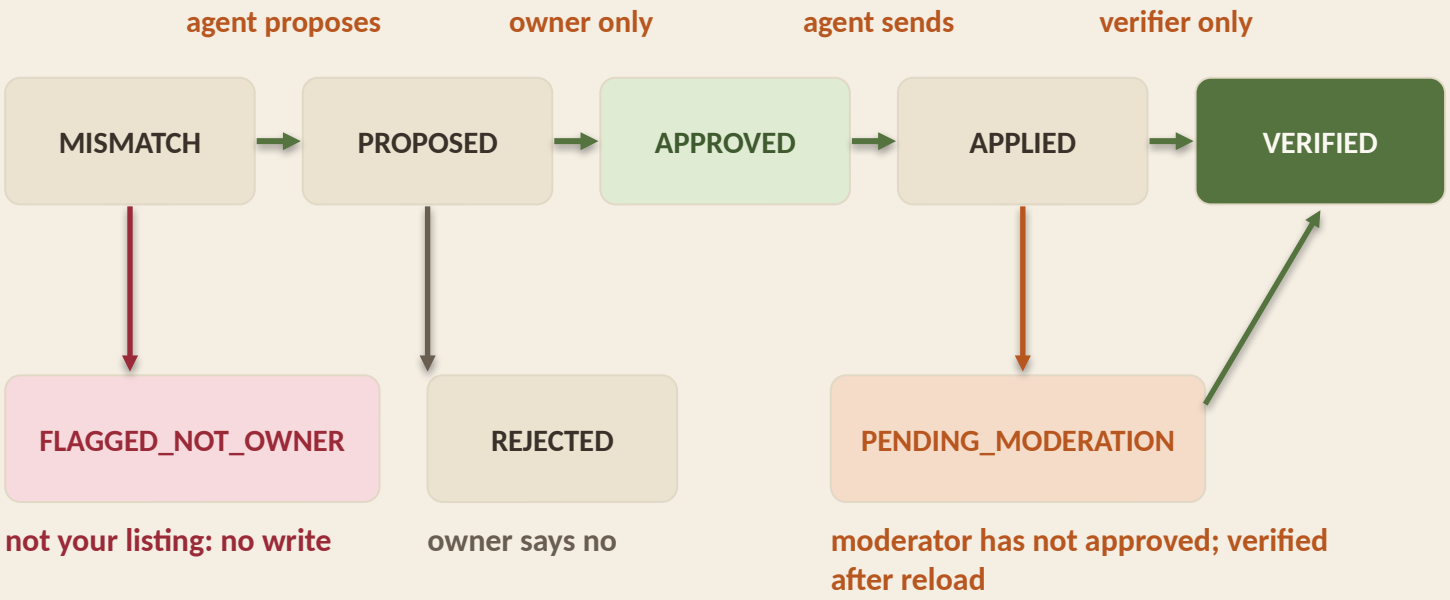
One service: console, agent, readers, five mock platforms

Data flow: record, read pages, normalise, compare, propose, owner approves, apply, reload and verify, report.



TypeScript · Node 22 · Express 5 · React 19 + Vite · SQLite · Zod 4 · Playwright + cheerio · chrono-node · Docker on Render

Safety is enforced in code, not in prompts



Only the owner can approve. Only the verifier can mark VERIFIED.

- One business can never see another's data
- PINs are never stored; the API key never leaves the server

Guardrails in code

- No apply without a DB row the owner approved
- Not your listing: flag, never edit
- Pending moderation is never reported as fixed
- Look-alike listings are never read or edited

6 / 6

unsafe actions blocked in the rogue-planner test: 0 writes

Not a dashboard, not a chatbot, not a cron job

	Dashboard	Chatbot	Rule-based sync	Our agent
Compares meaning, not formatting	No	Partly	No	Yes
Knows who owns a listing, refuses to edit it	Partly	No	No	Yes
Reloads the page to prove the edit is live	No	No	No	Yes
Tells "pending moderation" from "done"	No	No	No	Yes
Recovers when a page changes or a site is down	No	No	Partly	Yes

Typical tools in each category, not specific products. "Partly" means possible, or only if configured.

Verify-after-write

A field is VERIFIED only when the reloaded public page shows the approved value.

Finds you among look-alikes

Score = phone 40 + address 35 + name 25. The look-alike scores 37.7; the real listing 100.

Looks ahead, speaks plainly

Holiday alerts at 30, 14 and 7 days. "Closed on 26th December" becomes an approved change.

End to end: hours fixed, a stranger's listing left alone

1 — Scan and review

5 sites read in about 150 ms. CityPages shows Diwali as Closed; the record says 10:00-2:00. You decide.

2 — Safety

NearbyNow belongs to another account: zero edits, and a claim request is drafted.

3 — Apply and verify

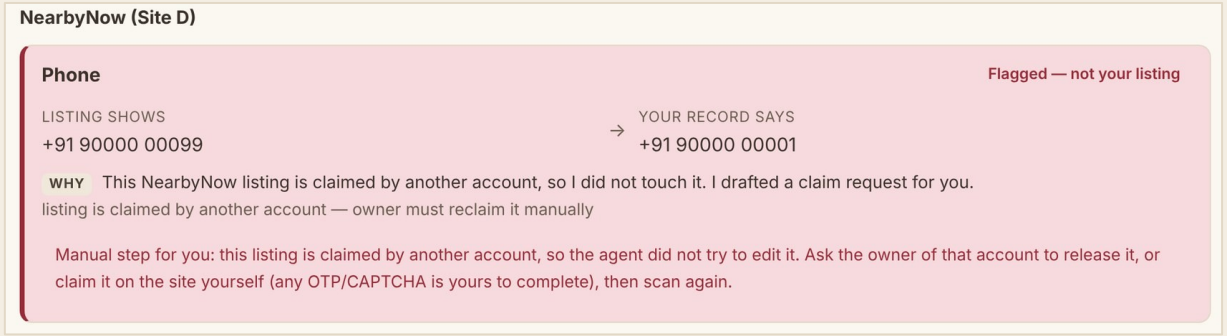
Only approved changes go out; pages are reloaded. ShopFront waits for its moderator, then a re-check verifies.

4 — Report

2 fixed and verified, 1 needs you, 2 ignored. Exported as CSV.



Review: the card shows both values and the reason.



Safety: not our listing, so nothing is edited.

TC1 to TC8 pass from a fresh reset (8 / 8) · 295 automated tests · 2-minute video: docs/demo/demo.mp4

Value we can measure, and where it goes next

~150 ms	3 clicks	8 / 8	0
full scan: 5 sites, 25 field checks	owner effort in the demo	test cases pass, plus 295 tests	unapproved writes in any test

Practical value

- No wrong-hours surprises on holidays
- Proof for the owner: VERIFIED, trace, CSV

Scalability

- Each platform is one more reader
- Every business is isolated by its own ID

Next

- Real platform APIs and consent flows
- Scheduled scans, PIN recovery, more fields

Honest limits mock platforms · OTP stays a human step · Gemini optional, not live-tested

Try it zirohacksub.onrender.com · demo PIN 7KQ4X-M9PHT · github.com/yourgrok/zirohacksub